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The Editors
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Dear Editors,

I am submitting for your consideration the manuscript “**An open-source GCR dosimetry pipeline for Mars mission risk assessment: organ-specific REID, SEP acute hazard, and global uncertainty quantification**” for publication in *Life Sciences in Space Research*.

What the paper does. The manuscript presents an open-source Python package (GCR-DOSIMETRY-PIPELINE) that integrates GCR spectrum generation, CSDA slab transport, organ-specific cancer risk (REID), Solar Energetic Particle acute dosimetry, and global uncertainty quantification via Latin Hypercube Sampling into a single reproducible workflow with 86 automated tests and a public repository.

What is new. I am transparent that the individual components — force-field modulation, CSDA transport, REID estimation — are established methods. The contributions are twofold.

First, no currently available open tool combines organ-specific REID, SEP acute risk, and global uncertainty propagation in a single workflow reproducible from a standard `pip install`, with inter-species flux ratios constrained to ACE/CRIS and PAMELA measurements rather than fit freely to a single total-dose calibration point.

Second, the Latin Hypercube Sampling ensemble shows that the radiobiological parameters (quality factor scale, DDREF, and the excess risk/absolute risk scaling factors) together account for approximately 88% of REID output variance, while GCR physics parameters contribute approximately 12%, quantifying why physics improvements alone cannot close mission risk uncertainty. This is a directly actionable finding for research prioritization.

Scope and limitations. The transport model uses CSDA with exponential nuclear attenuation and does not capture heavy-ion fragmentation secondaries. This limitation is stated explicitly and its domain of validity is discussed. The per-species calibration to MSL/RAD is described as calibration, not independent prediction, and the validation evidence is presented accordingly. The REID framework follows Cucinotta (2013); the 2023 NASA update is noted as a limitation for future work. The repository also contains a separate LET/RBE proton-therapy benchmarking layer, reported here only as a brief implementation self-consistency check rather than a central contribution, since it is the foundation of a distinct follow-up

study.

Relevance to *Life Sciences in Space Research*. This manuscript is squarely within the journal's scope: human spaceflight radiation risk, Mars mission planning, and space radiation biology.

Data and code availability. All source code, input data, and validation scripts are openly available on GitHub at the repository [aryanhshah8/gcr-dosimetry-pipeline](https://github.com/aryanhshah8/gcr-dosimetry-pipeline), and no proprietary datasets are required to reproduce any results.

I confirm that this manuscript has not been submitted elsewhere, that I am the sole author, and that I have no conflicts of interest to declare. I look forward to the reviewers' comments.

Sincerely,

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